

Unit ID: 6739dae0a34577ff4ce14950

Unit 2 Bringing Ideas to Life Building a 4-level mood lamp using Modi Modules

PDF Documents

PDF Document 1: 1731844832582-Chapter 6_ unit2 Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

Update #1 - 2025-05-16 15:14:04



UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

STORY TELLING - ASHTON'S MOOD LAMP



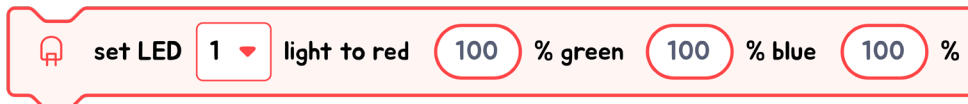
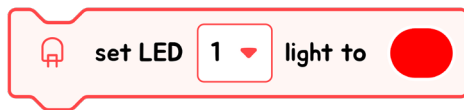
In Mr. Thompson's sixth-grade class at Lux Elementary, Ashton was buzzing with excitement for their latest project. The task? Build a 4-Level Mood Lamp using the MODI kit, with each color change—red, orange, blue, and white—representing a different temperature of fire.

As Ashton and his classmates assembled their lamps, he marveled at how the colors shifted with each button press, each hue demonstrating a warmer stage of fire. The project not only highlighted the relationship between temperature and color but also brought the abstract concept of heat to vibrant, visual life.

When the classroom lights dimmed and the students' mood lamps illuminated the room, Ashton felt a surge of pride. His lamp worked flawlessly, cycling through colors that painted the classroom in a warm glow.



ABOUT LED MODULE

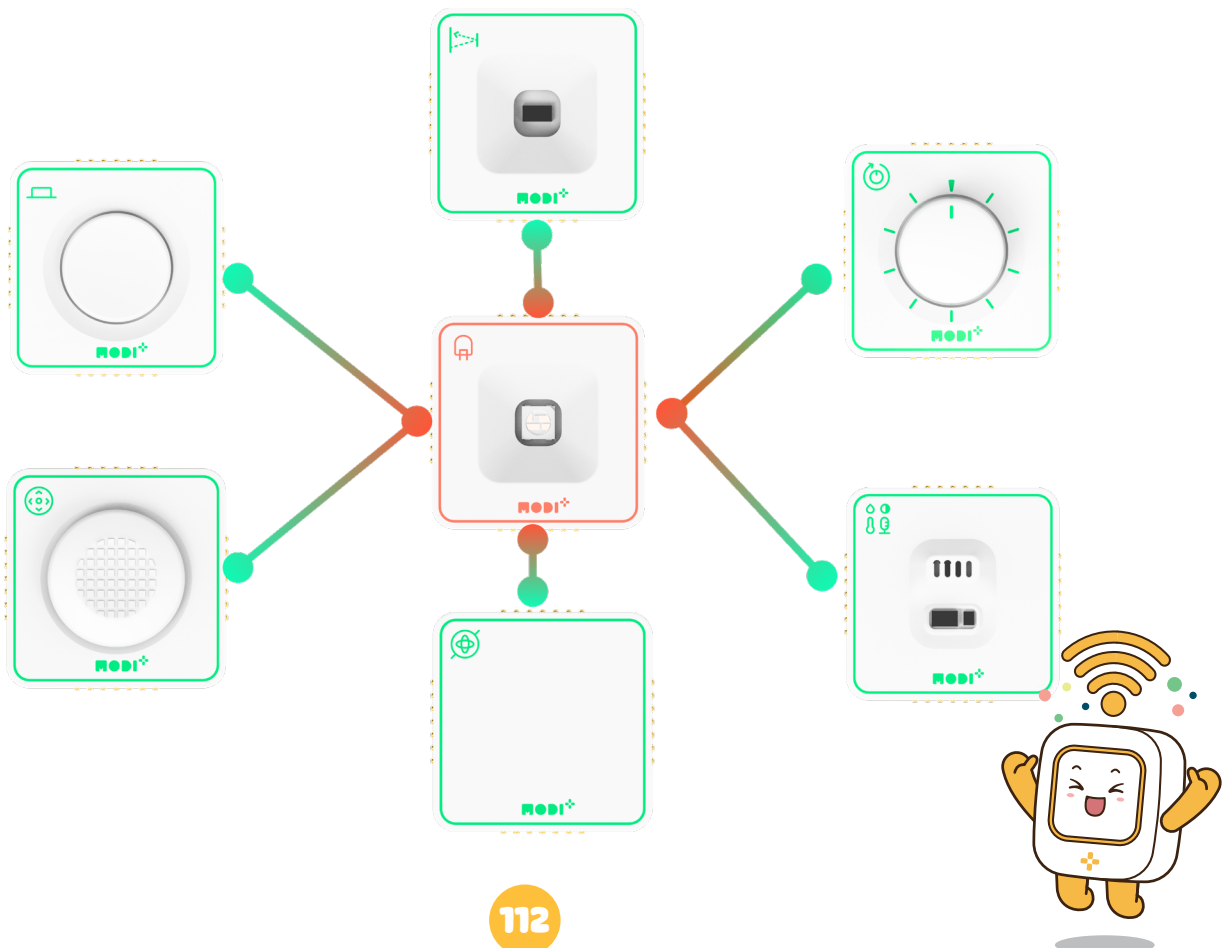


The LED module features three distinct blocks in the block area. The first block enables you to select a color using hue, saturation, and brightness values. The second block allows you to set the color based on RGB (Red, Green, Blue) values. The third block is used to turn off the LED module

INTERACTIONS WITH OTHER MODULES

As an output module, the LED can interact with all input modules. You can observe these interactions by connecting the LED module to any input module, even without any programming required!

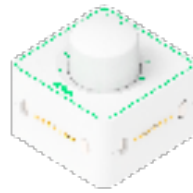
INTERACTION BETWEEN LED AND OTHER MODULES





Required Modules & Brainstorming

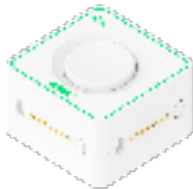
Based on the story, which MODI modules do you need for this creation?



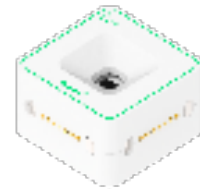
Dial



Environment



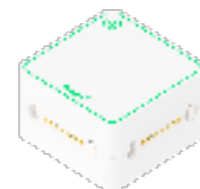
Button



ToF



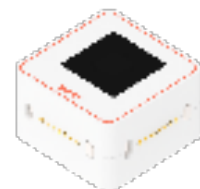
Joystick



IMU



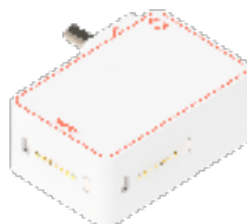
Speaker



Display



LED

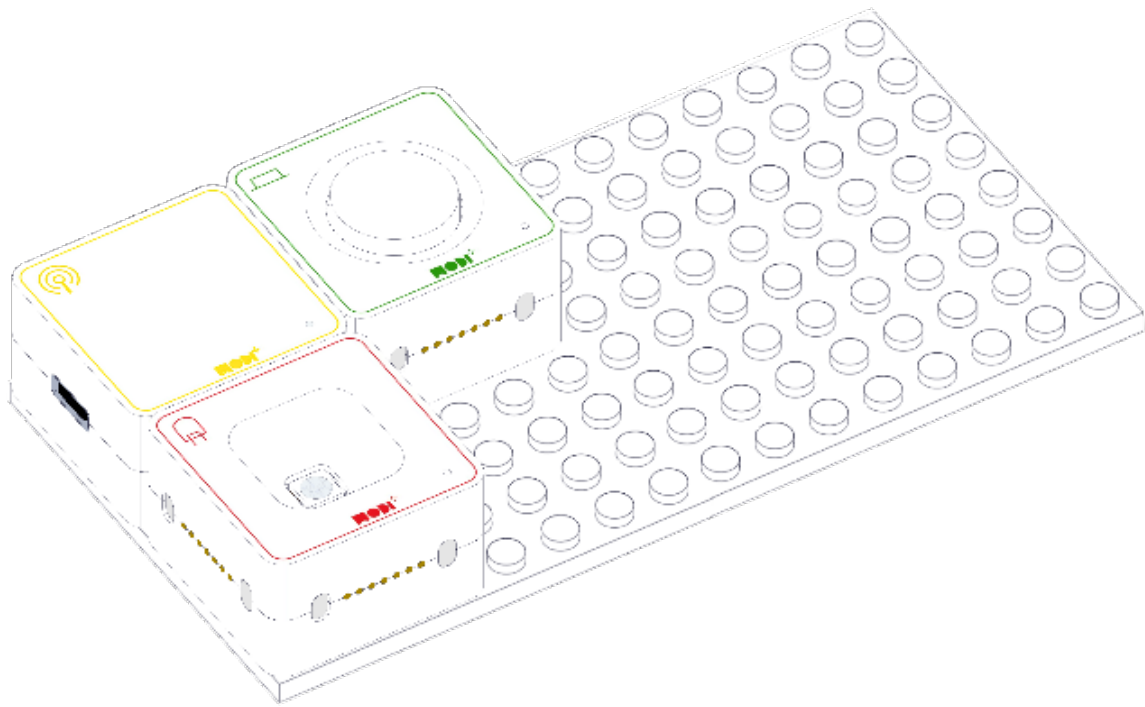


Motor A



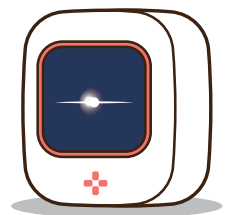
Motor B

HOW DOES THE CREATION LOOK LIKE?

**TIPS**

Want to make your mood lamp even cooler?

Try using colorful or patterned paper to create a unique lampshade. Make your lamp not only a science experiment but also a cool decor piece for your room!





Module Functionality

Let's look detail about the Button module.



- ✓ clicked
- double clicked
- pressed
- toggle on

Button can receive 4 types of information.

1. Click



Detects if Button is clicked or not.
Range: True(Click once) : 100 / False : 0

2. Double click



Detects if Button is double - clicked or not.
Range: True(Click twice) : 100 / False : 0

3. Press



Show the press status of Button.
Range: True(Pressed) : 100 / False : 0

4. Toggle



Changes on / off when Button is clicked.
Range: True(On) : 100 / False : 0

THINK ABOUT IT!

Do you remember which action Button can receive?

